

ALPACA AI TRADING AGENTS HACKATHON

AlgoSentinel

Autonomous AI-Powered Options Trading Agent

Team AlgoSentinel

Paper Trading Account • Real-Time News Sentiment • SPY Options Execution

The Problem & Our Solution

THE PROBLEM

Most traders are emotional, biased, and slow.

Opportunities in market sentiment die in seconds.

- Human reaction latency misses fast-moving market catalyst headlines.
- Cognitive bias & panic trading lead to broken discipline and blown accounts.
- Manually screening options contracts, strikes, and ask prices takes too long.

OUR SOLUTION

→ **AI reads live news in real-time**

Pulls S&P 500 & SPY RSS feeds every 15 minutes

→ **Scores sentiment automatically**

Gemini AI parses tone, nuances, and macro implications

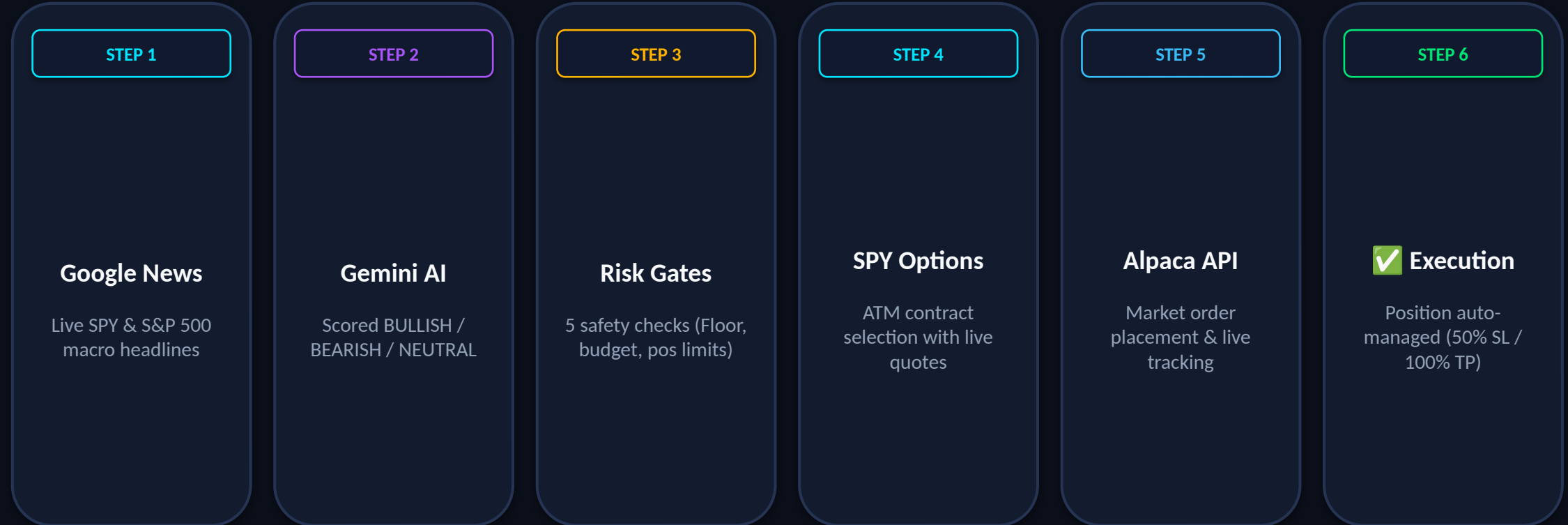
→ **Executes trades instantly**

Direct Alpaca Trading API execution with optimal ATM strikes

→ **No human delays. No emotion.**

Pure signal-driven trading governed by strict risk gates

How It Works: End-to-End Pipeline



 Every 15 minutes. Fully autonomous. Zero human intervention needed.

Why It Matters: Institutional Discipline for AI



AUTONOMOUS & AI-DRIVEN

- **Zero Human Latency**
Runs every 15 mins during market hours
- **Gemini Sentiment Engine**
Gemini scores sentiment & drives decisions
- **Complete Audit Trail**
Every trade and skip logged to trades.log



EXECUTION & RESULTS

- **Fresh \$100k Paper Account**
Live paper trading via Alpaca API
- **Intelligent Position Exits**
-50% Stop-loss & +100% Take-profit protection
- **Hackathon Stack**
Alpaca Trading API • Google Gemini • Python



5-TIER RISK MANAGEMENT

- ✓ **Equity Floor**
\$85k minimum capital floor
- ✓ **Max 3 Positions**
Strict limit on simultaneous exposure
- ✓ **5% Per-Trade Cap**
Max budget per individual contract
- ✓ **Confidence Filter**
Only trade if signal confidence $\geq 60\%$
- ✓ **Market Hours Only**
Active 9:30 AM – 4:00 PM ET